Kindergarten Ontario Curriculum Number / Coding

Little Programmers — Kindergarten Coding

Unit Overview: Little Programmers — Kindergarten Coding

Grade: Kindergarten | Strand: Number / Coding

Target Expectations: B11.1, B11.2, B11.3

Learning Goal: Children will use directional and positional words, record step-by-step instructions, and test/refine them.

Lesson	Lesson Title	Learning Goal
Day 1	**Move with Bot**	I can use direction words to move.
Day 2	**Where is Bot?**	I can use position words.
Day 3	**Code with Pictures**	I can record steps with pictures.
Day 4	**Find the Bug**	I can test and refine my steps.
Day 5	**The Big Programmer Capstone**	I can move, record, and test instructions.

Lesson 1: Move with Bot

Goal: DIRECTIONAL words tell us HOW TO MOVE: forward, back, left, right.

Expectation: B11.1

Learning Goal: "I can use direction words to move."

Materials: M1_floor_mat, M2_arrow_cards, M7_coding_chart, M10_unit_poster

1. Minds On (10 mins): Bot's First Move

Bot the Robot Programmer: BEEP BOOP! Friends, I move with DIRECTION words. Forward! Back! Left! Right! Bot opens: 'STEP, STEP, STEP — that's CODE!'

Bring children to the carpet. Bot the Robot puppet (silver foil over square box). 'Coder friends — I'm Bot the Robot. I MOVE with FOUR direction words. Watch!'

2. Action (20 mins): Move Bot 8 Steps

- Hand out Floor Mats + Arrow Cards.
- Pairs trade: one calls, one moves.
- Stretch: include 1 LEFT and 1 RIGHT in the 4 steps.
- Help with left vs right (use stickers on hands).
- Circulate to record diagnostic.
- INVENT-AND-COMPARE: Two pairs share 4-step routes to the SAME target.

3. Consolidation (10 mins): Bot's Move Brag

Discuss:
- Name 4 direction words.
- Show FORWARD with your body.
- Which is your LEFT hand?
- What if Bot has NO direction words?
- Bot's preview: tomorrow Bot HIDES and we have to find HIM (Lesson 2)!

Exit: Bot's refrain — 'Bot says: STEP, STEP, STEP — that's CODE!' (3x).

Lesson 2: Where is Bot?

Goal: POSITIONAL words tell us WHERE: above, below, beside, between.

Expectation: B11.1

Learning Goal: "I can use position words."

Materials: M1_floor_mat, M3_position_chart, M4_object_cards, M7_coding_chart

1. Minds On (10 mins): Bot's Hide and Seek

Bot the Robot: Friends, I'm hiding! ABOVE? BELOW? BESIDE? BETWEEN? Find me with POSITION words! Bot opens: 'STEP, STEP, STEP — that's CODE!'

Bring children to the carpet. Bot the Robot puppet. 'Coder friends — yesterday we MOVED. Today we tell WHERE with POSITION words.'

2. Action (20 mins): Describe Bot's Spot 4 Times

- Hand out Floor Mat + 4 Object Cards.
- Hide Bot 4 different spots.
- Stretch: use 2 position words for one spot.
- Help with above vs below (point up/down).
- Circulate to record formative.
- STRATEGY COMPARE: After 4 spots, two pairs share HOW they decided ABOVE/BELOW/LEFT/RIGHT — by Bot's view, by their own view, by absolute room directions.

3. Consolidation (10 mins): Bot's Position Brag

Discuss:

- Name 4 position words.
- ABOVE or BELOW your hand?
- Show BETWEEN.
- Where is your pencil right now?

Exit: Bot's refrain (carry-over): 'STEP, STEP, STEP — that's CODE!' (3x).

Lesson 3: Code with Pictures

Goal: We RECORD steps with arrows and pictures so anyone can follow.

Expectation: B11.3

Learning Goal: "I can record steps with pictures."

Materials: M1_floor_mat, M2_arrow_cards, M4_object_cards, M5_code_strips

1. Minds On (10 mins): Bot's Picture Recipe

Bot the Robot: To MOVE me, you need a CODE. Like a recipe — pictures of each step! Bot opens: 'WRITE the arrows, BOT will GO!' Day 1 we MOVED, Day 2 we FOUND, today we WRITE.

Bring children to the carpet. Bot the Robot puppet. 'Coder friends — yesterday I MOVED with words. But what if my friend isn't here? I need to RECORD my code!'

2. Action (20 mins): Record 2 Codes

- Hand out Code Strips.
- Plan and record a 4-step code to reach apple.
- Trade with another pair.
- Help with arrow direction (always read left-to-right).
- Circulate to record formative.

3. Consolidation (10 mins): Bot's Code Brag

Discuss:

- What's a CODE?
- Why RECORD steps?
- How do you READ a code?
- How is code like a recipe?

Exit: Bot's coding refrain — 'WRITE the arrows, BOT will GO!' (3x).

Lesson 4: Find the Bug

Goal: TEST your code. If wrong, REFINE — find the BUG and fix it!

Expectation: B11.2

Learning Goal: "I can test and refine my steps."

Materials: M1_floor_mat, M5_code_strips, M8_bug_cards, M7_coding_chart

1. Minds On (10 mins): Buggy's Bug Hunt

Buggy the Beetle: BZZ! Friends, sometimes code has a BUG — a mistake! TEST first. Then FIX! Buggy opens: 'TEST it, FIX it, TEST it AGAIN!' Yesterday on Day 3 we RECORDED codes with arrows. Today we HUNT bugs in them.

Bring children to the carpet. Buggy the Beetle puppet (small green felt with magnifying glass). 'Coder friends — meet Buggy! Buggy finds BUGS in code!'

2. Action (20 mins): Fix 4 Buggy Codes

• Hand out 4 Bug Cards.
• Find the BUG and write the FIXED code.
• Stretch: explain WHERE the bug was.
• Help by walking the code on a tape mat.
• Circulate to record formative.
• STRATEGY COMPARE: After step 2, prompt 'How did you test? Walk it, trace with finger, mental?' Two pairs share.

3. Consolidation (10 mins): Buggy's Bug Brag

Discuss:
- What's a BUG?
- What does TEST mean?
- What's REFINE?
- Why are bugs OK?

Exit: Bot+Buggy debug refrain — 'TEST it, FIX it, TEST it AGAIN!' (3x).

Lesson 5: The Big Programmer Capstone

Goal: Programmers MOVE, RECORD, and TEST step by step.

Expectation: B11.1, B11.2, B11.3

Learning Goal: "I can move, record, and test instructions."

Materials: M1_floor_mat, M2_arrow_cards, M3_position_chart, M4_object_cards

1. Minds On (10 mins): Bot's Big Programmer Day

Bot the Robot: Today's BIG case — move, position, record, test! All 4 in ONE day! Bot+Buggy open: 'MOVE it, RECORD it, FIND the BUG — that's CODING!'

Bring children to the carpet. Bot and Buggy puppets in capes. 'Coder friends — today is our CAPSTONE. We do ALL we know: directions, positions, codes, and bug fixing!'

2. Action (20 mins): The Big Programmer

- Hand out templates.
- Stations 1-4: move, position, record, test.
- Stretch: write a 'programmer's brag' — one sentence per station.
- Help with bug fixing at Station 4.
- Circulate to record summative evidence.
- STRATEGY COMPARE at Station 3 or in consolidation: Two children share their 4-step codes.

3. Consolidation (10 mins): Bot and Buggy's Big Brag

Discuss:

- What are 4 direction words?
- What are 4 position words?
- What's a CODE?
- What's a BUG?

Exit: Capstone refrain — Bot+Buggy: 'MOVE it, RECORD it, FIND the BUG — that's CODING!' (3x).

Name: ____________________ Date: ____________________

Bot's First Moves

Learning Goal: I can use direction words to move.

Instructions for the Student:

1. Draw a line from each direction word to its matching arrow.
2. Bot follows: forward, forward, right, forward.
3. Draw an arrow showing Bot how to go to the apple.

Part 2: Colour the path.

Bot follows: forward, forward, right, forward. Colour Bot's path on the mat.

(3 parts on the printable worksheet)

Name: ______________________________ Date: ____________________________

Where is Bot?

Learning Goal: I can use position words.

Instructions for the Student:

1. Draw a line from each position word to the matching picture.
2. For 4 pictures, circle the right position word: ABOVE, BELOW, BESIDE, or BETWEEN.
3. Draw Bot ABOVE the star.

Part 2: Circle the position.

For 4 pictures, circle the right position word: ABOVE, BELOW, BESIDE, or BETWEEN.

(3 parts on the printable worksheet)

Name: ______________________ Date: ______________________

Bot's Picture Code

Learning Goal: I can record steps with pictures.

Instructions for the Student:

1. Three codes show paths.
2. Draw a 4-step code on the strip to reach the heart.
3. Number 4 scrambled arrows in order so Bot reaches the star.

Part 1: Pick the right code.

Three codes show paths. Circle the one that reaches the apple.

(3 parts on the printable worksheet)

Name: ______________________ Date: ______________________

Buggy's Bug Hunt

Learning Goal: I can test and refine my steps.

Instructions for the Student:

1. For 3 codes, circle BUG or NO BUG.
2. One code has a BUG.
3. Write a 4-step code that reaches the apple.

Part 3: Write a no-bug code.

Write a 4-step code that reaches the apple. Make sure NO BUG!

(3 parts on the printable worksheet)

Name: ____________________ Date: ____________________

Bot and Buggy Capstone

Learning Goal: I can move, record, and test instructions.

Instructions for the Student:

1. Match 4 direction words to 4 arrows.
2. Look at 2 pictures.
3. Write a 4-step code to reach the apple.

Part 2: Position word.

Look at 2 pictures. Write the position word: above, below, beside, or between.

(3 parts on the printable worksheet)

Coding Floor Mat

(M1_floor_mat)

Letter-size 4x4 grid mat for Bot to move on.

Coding Floor Mat

	A	B	C	D
1				
2				
3				
4				

PRINT SPECS
- Size: letter landscape
- Colour: bw
- Pages per set: 1
- Laminate: yes

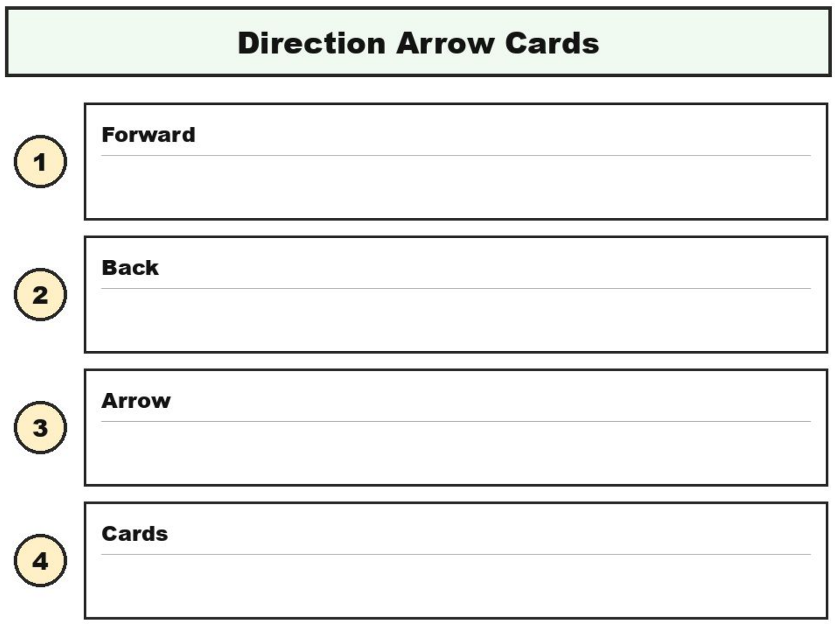
USE IN CLASS
- Quantity: 10-12 mats
- Lessons: 1, 2, 3, 4, 5
- Prep: ~15 min

TEACHER PREP
1. Print 12 mats.
2. Laminate.
3. Distribute one per pair.

Direction Arrow Cards

(M2_arrow_cards)

20 paper arrow cards (5 each direction).



PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS
Quantity: 10 sets
Lessons: 1, 3, 5
Prep: ~25 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 20 cards per set.
4. Store in labelled bag.

Position Words Chart

(M3_position_chart)

Anchor chart with position word icons.

Position Words

LEFT	RIGHT	UP	DOWN
ABOVE	BELOW	INSIDE	OUTSIDE

PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

USE IN CLASS
- Quantity: 1 chart
- Lessons: 2, 5
- Prep: ~10 min

TEACHER PREP
1. Print 1 copy.
2. Laminate.
3. Tape to front wall before Lesson 2.

Object Cards (Bot Targets)

(M4_object_cards)

12 paper object targets for Bot to navigate to.

PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

USE IN CLASS
- Quantity: 10 sets
- Lessons: 2, 3, 5
- Prep: ~20 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 12 cards per set.
4. Store in labelled bag.

Step-by-Step Code Strips

(M5_code_strips)

20 blank paper strips for recording 4-step routes.

Step-by-Step Code Strips

PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: no

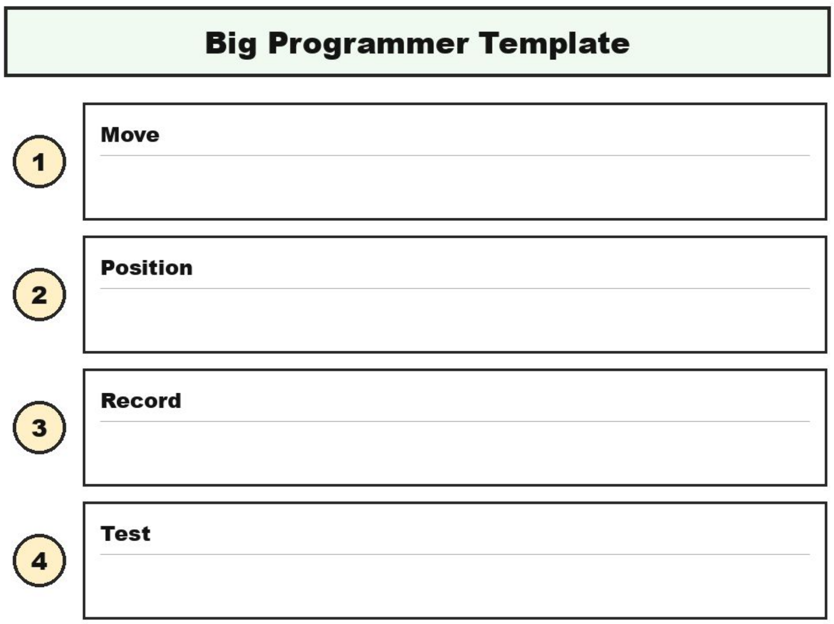
USE IN CLASS
- Quantity: 5 sheets
- Lessons: 3, 4, 5
- Prep: ~15 min

TEACHER PREP
1. Print 5 sheets.
2. Cut into 20 strips per sheet.
3. Stack near teacher table.

Big Programmer Template

(M6_capstone_template)

Letter-size capstone template with 4 sections.



PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: no

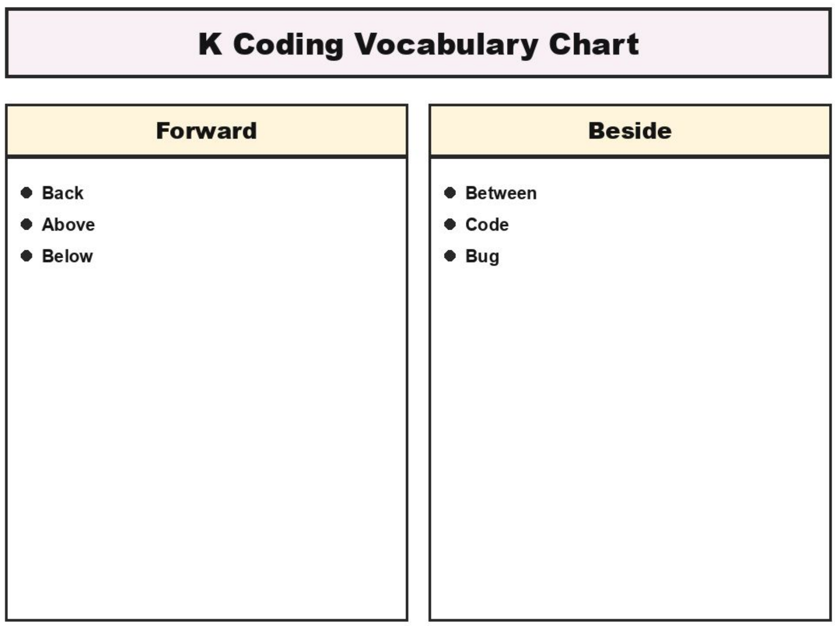
USE IN CLASS
- Quantity: 25 templates
- Lessons: 5
- Prep: ~5 min

TEACHER PREP
1. Print 25 templates.
2. Stack near teacher table.
3. No lamination.

K Coding Vocabulary Chart

(M7_coding_chart)

Anchor chart with K coding vocabulary.



PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

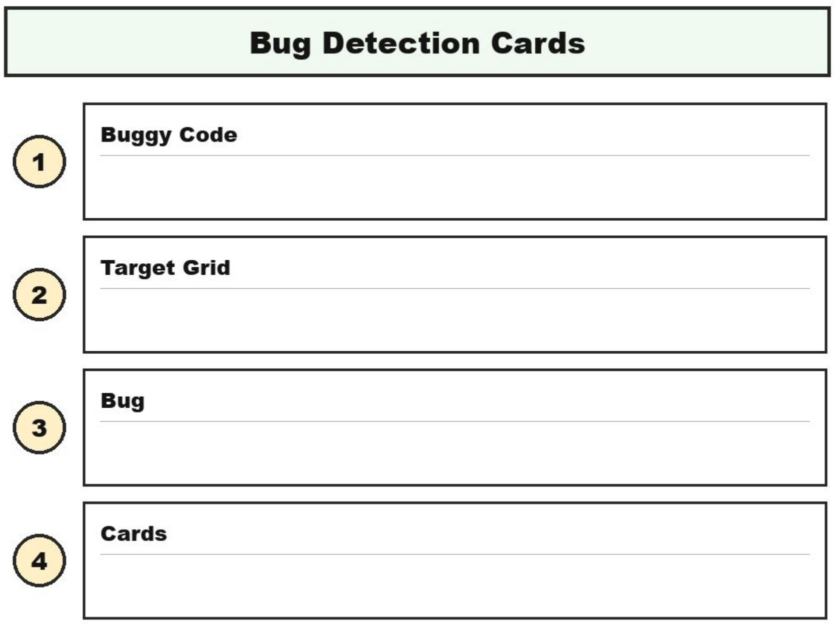
USE IN CLASS
Quantity: 1 chart
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

TEACHER PREP
1. Print 1 copy.
2. Laminate.
3. Tape to front wall before Lesson 1.

Bug Detection Cards

(M8_bug_cards)

12 broken-code cards for debugging practice.



PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

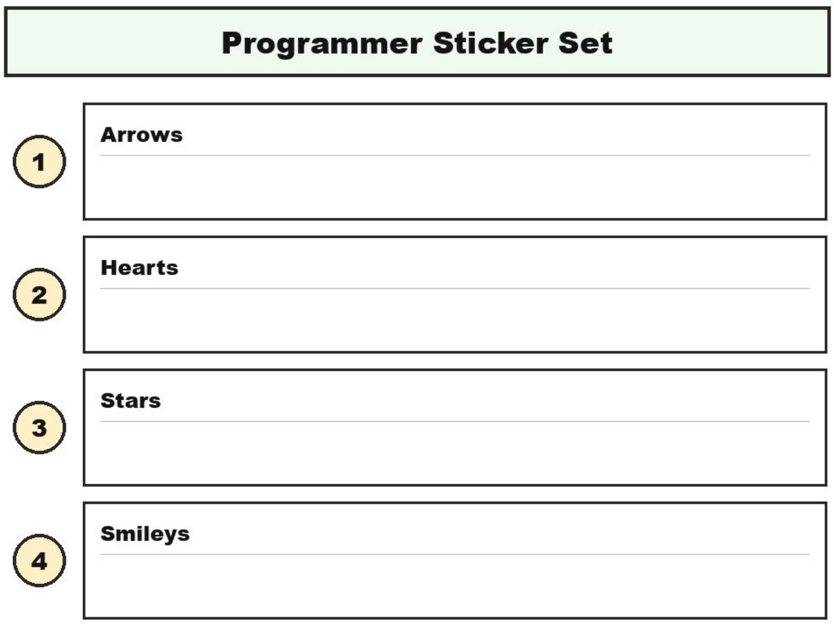
USE IN CLASS
- Quantity: 10 sets
- Lessons: 4, 5
- Prep: ~25 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 12 cards per set.
4. Store in labelled bag.

Programmer Sticker Set

(M9_sticker_set)

30 mini stickers for celebrations.



PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: no

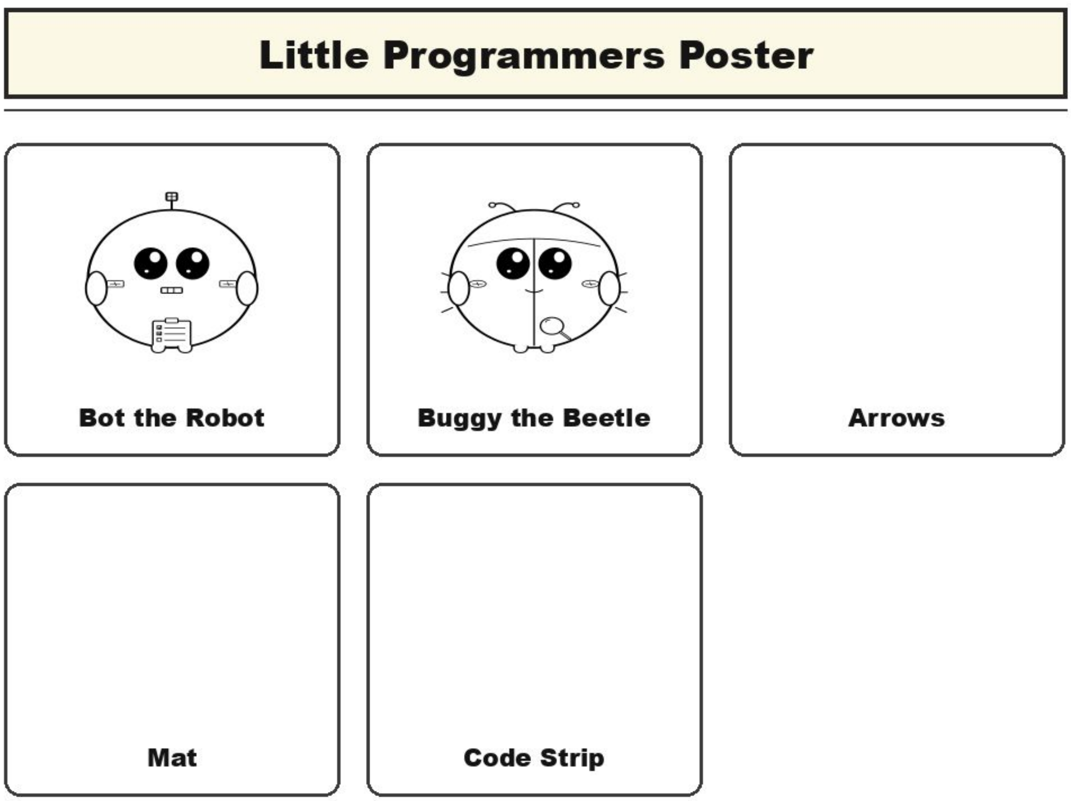
USE IN CLASS
Quantity: 5 sheets
Lessons: 5
Prep: ~15 min

TEACHER PREP
1. Print 5 sheets.
2. Cut into 30 stickers per sheet.
3. Use peel-and-stick paper if available.

Little Programmers Poster

(M10_unit_poster)

Unit anchor poster of Bot and Buggy.



PRINT SPECS
- Size: letter landscape
- Colour: bw
- Pages per set: 1
- Laminate: yes

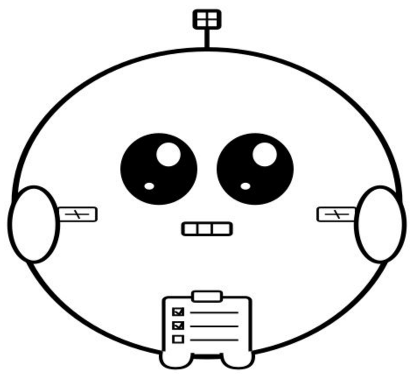
USE IN CLASS
- Quantity: 1 poster
- Lessons: 1, 2, 3, 4, 5
- Prep: ~10 min

TEACHER PREP
1. Print 1 copy on letter or 11x17.
2. Laminate.
3. Tape to front wall on Day 1.

Bot Puppet

(char_bot_puppet)

Paper puppet of Bot the Robot for teacher use in all 5 lessons.



Bot the Robot Programmer

PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

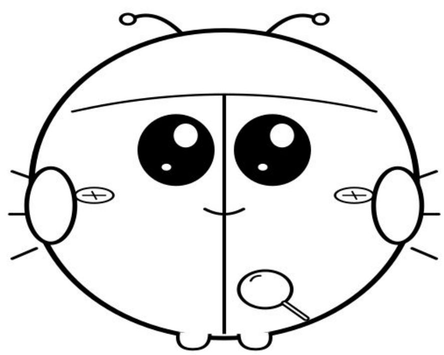
USE IN CLASS
- Quantity: 1 puppet
- Lessons: 1, 2, 3, 4, 5
- Prep: ~12 min

TEACHER PREP
1. Print 1 copy.
2. Cut around silhouette and tab.
3. Laminate.
4. Glue popsicle stick.

Buggy Puppet

(char_buggy_puppet)

Paper puppet of Buggy the Debug Beetle for teacher use in Lessons 4-5.



Buggy the Debug Beetle

PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS
Quantity: 1 puppet
Lessons: 4, 5
Prep: ~12 min

TEACHER PREP
1. Print 1 copy.
2. Cut around silhouette and tab.
3. Laminate.
4. Glue popsicle stick.

Name: ______________________ Date: ______________________

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
B11.1 use directional and positional language to create and follow instructions involving movement	Cannot name directions. Mixes left and right.	Uses 1-2 direction or position words with prompts.	Independently uses 4 direction and 4 position words to create and follow instructions.	Combines direction and position words in multistep instructions.
B11.2 test and refine instructions	Does not test code. Says 'no bug' without checking.	Tests step-by-step with prompts.	Independently tests instructions and finds and fixes bugs.	Articulates the bug location and proposes multiple fixes.
B11.3 communicate and record step-by-step instructions using symbols, words, or pictures	Cannot record arrows. Random order.	Records 1-2 steps with prompts.	Independently records a 4-step code in left-to-right order.	Designs original codes for a partner to follow.

Junior Programmer Certificate (K)

This certificate is proudly presented to:

____

For successfully completing the 5-Day K Little Programmers unit and demonstrating the skills of a K Junior Programmer. Bot the Robot congratulates the student.

By completing this unit, this student has demonstrated:

- Using direction words (forward, back, left, right)
- Using position words (above, below, beside, between)
- Recording step-by-step codes with arrows
- Testing code and finding bugs

Marketplace Listing

A 5-day Kindergarten unit on foundational coding framed around Bot the Robot Programmer and Buggy the Debug Beetle.

Price: $7.99 CAD
Pages: ~40
Classroom time: 200 min
Teacher prep: ~180 min

Top tags: Kindergarten, math, coding, robot, directions, position, debugging, Ontario curriculum

What's included:

- Unit blueprint (1 file)
- 5 lesson plans (5 files)
- 5 worksheets (5 files)
- Manipulatives plan (1 file)
- Formative + reflection (1 file)
- Assessment suite (1 file)